Table 1: S. cerevisiae datasets (pre-build, 2026-10-15) built as LMDBs in the TorchCell dev tree, with a description of each. *Description* is written from the loader module and its note. *Sequence-only fit* and *Label unit* are derived from the genotype of 400 evenly spaced records per store: how well a sequence-embedding-only model fits the dataset, and what one label attaches to without the cell graph. The other columns are as in the non-expanded table.

Dataset	Host	Description	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bits)	Sequence-only fit	Label unit
Environmental / chemogenomic response											
CellYieldDeChiara2022	*S. cerevisiae*	De Chiara 2022 grew the 1,011 sequenced isolates of the Peter 2018 panel as colonies on 43 carbon and nitrogen environments, reporting growth from Supplementary Table 3. This store holds the absolute colony cell yield. Each isolate is a perturbation set off S288C, and one record is one isolate in one environment.	906	43	36,286	environment_response (cell_yield)	scalar	global	1.7×10^{7}	no	natural genome (segregant or isolate)
CrisprMagicLian2019	*S. cerevisiae*	Lian 2019 built MAGIC, genome-scale CRISPRa, CRISPRi and CRISPRd guide libraries run by three orthogonal Cas effectors in one pooled host, and selected them in SED-URA/G418 with 5, 10 or 15 mM furfural over three iterative rounds. One record is one guide in one round: its mean log2 enrichment over three biological replicates.	266,304	3	266,304	environment_response (log2_ratio)	scalar	global	1.3×10^{8}	good	1 gene x condition (1-3 edits)
CrispriChemgenSmith2016	*S. cerevisiae*	Smith 2016 grew pooled dCas9-Mxi1 CRISPRi gRNA tiling libraries against 20 drug-partner genes in BY4741 in competitive growth, with and without ATc and with or without a small-molecule inhibitor, tracking guides by amplicon sequencing. One record is one pool, guide and drug condition: the ATc-induced log2 fold change and its variance.	1,035	26	14,463	environment_response (log2_ratio)	scalar	global	7.5×10^{6}	good	1 gene x condition
CrispriMormino2022	*S. cerevisiae*	Mormino 2022 integrated a Haa1-based acetic acid biosensor into a pooled dCas9-Mxi1 CRISPRi library and FACS-enriched cells with the highest reporter signal in SC with 50 mM acetic acid at pH 3.5. One record is one of 12 isolated strains with one CRISPRi target and its categorical biosensor call against the library pool.	12	1	12	environment_response (categorical)	scalar	global	1.4×10^{4}	good	gene triple, strain readout
DoublingTimeDeChiara2022	*S. cerevisiae*	De Chiara 2022 grew the 1,011 sequenced isolates of the Peter 2018 panel as colonies on 43 carbon and nitrogen environments, reporting growth from Supplementary Table 3. This store holds the absolute colony population doubling time in hours. Each isolate is a perturbation set off S288C, and one record is one isolate in one environment.	906	43	36,295	environment_response (growth_rate)	scalar	global	1.7×10^{7}	no	natural genome (segregant or isolate)
EnvChemgenAuesukaree2009	*S. cerevisiae*	Auesukaree 2009 screened the 4,828-strain haploid BY4742 KanMX deletion collection by serial-dilution spot assay on YPD plates for sensitivity to six stresses: ethanol, methanol, 1-propanol, 37 C heat, 1 M NaCl and 5 mM H2O2. Only sensitive hits are reported, so one record is one sensitive deletion x stress pair with a categorical call.	333	6	525	environment_response (categorical)	scalar	global	3.1×10^{5}	good	1 gene x condition
EnvChemgenCostanzo2021	*S. cerevisiae*	Costanzo 2021 ran condition SGA screens of deletion and temperature-sensitive allele arrays in 14 environmental conditions (13 added compounds and galactose as an alternative carbon source) plus a matched reference condition, scoring colony size. One record is one mutant in one condition: its differential fitness against the reference condition.	4,422	14	61,640	environment_response (differential_fitness)	scalar	global	1.1×10^{8}	good	1 gene x condition
EnvChemgenHoepfner2014	*S. cerevisiae*	Hoepfner 2014 profiled the heterozygous (HIP) and homozygous (HOP) kanMX4 deletion collections against 1,776 compounds, each in duplicate near its IC30, mostly proprietary Novartis CMB compounds. One record is one deletion strain under one compound dose and study, with its sensitivity score; 30,308,574 records.	10,839	5,586	30,308,574	environment_response (sensitivity_score)	scalar	global	2.1×10^{10}	good	1 gene x condition
EnvChemgenMota2024	*S. cerevisiae*	Mota 2024 screened the haploid BY4741 Euroscarf KanMX deletion collection by spot assay on YPD agar at pH 4.5 and 30 C against 75 mM acetic, 14 mM butyric and 0.30 mM octanoic acid, scored against the parental strain. One record is one deletion x acid pair with an ordinal susceptibility grade (0, +, ++).	607	3	1,289	environment_response (ordinal)	scalar	global	7.4×10^{5}	good	1 gene x condition
EnvChemgenParsons2006	*S. cerevisiae*	Parsons 2006 grew the pooled MATa haploid deletion collection against 82 conditions (75 compounds and 7 crude natural-product extracts) and read strain abundance from uptag and downtag barcodes on Agilent arrays. One record is one of 4,111 deletion strains in one condition, with the released log2(control/treated) ratio, positive when sensitive.	4,111	82	329,254	environment_response (log2_ratio)	scalar	global	2.2×10^{8}	good	1 gene x condition
EnvChemgenPiotrowski2017	*S. cerevisiae*	Piotrowski 2017 profiled compound conditions from seven libraries against a 310-strain diagnostic pool of kanMX deletions in a pdr1 pdr3 snq2 drug-sensitized background, grown 48 h at 30 C in YP galactose and read by UPTAG barcode sequencing. One record is one strain in one condition with its chemical-genetic interaction z-score.	289	11,949	3,435,820	environment_response (z_score)	scalar	global	1.4×10^{9}	good	1 gene x condition
EnvChemgenVanacloig2022	*S. cerevisiae*	Vanacloig-Pedros 2022 grew a drug-sensitized barcoded yeast deletion library anaerobically against 45 released conditions, mostly plant-hydrolysate inhibitors, in biological triplicate beside paired inhibitor-free controls, counting barcodes. One record is one deletion in one condition: the mean log2(inhibitor/control) of TMM-normalized counts with its SD.	3,649	45	164,205	environment_response (log2_ratio)	scalar	global	9.3×10^{7}	good	1 gene x condition
EnvChemgenWildenhain2015	*S. cerevisiae*	Wildenhain 2015 screened compounds at 20 uM against sentinel BY4741 kanMX4 deletion strains, scoring growth inhibition by OD600; the released PubChem extension covers 5,518 compounds and 242 strains in duplicate. One record is one released screen of one strain against one compound, with its z-score against that strain's plate median.	258	5,178	449,033	environment_response (z_score)	scalar	global	2.6×10^{8}	good	1 gene x condition (0-1 edits)
EnvChemgenYoshikawa2009	*S. cerevisiae*	Yoshikawa 2009 grew 4,729 MATalpha single-gene deletion strains one per well in static 96-well YPD at 30 C, adding water, ethanol (8% final) or NaCl (1 M final), and fitted specific growth rate from OD. Each strain was cultured twice per condition, and one record is one strain, condition and culture with its normalized growth rate.	4,729	4	37,832	environment_response (growth_rate)	scalar	global	1.0×10^{7}	good	1 gene x condition
FattyAcidSmith2006	*S. cerevisiae*	Smith 2006 screened the MATalpha haploid BY4742 KanMX deletion set on agar for peroxisomal fatty acid beta-oxidation: clear-zone size on oleate (YPBO) and myristate (YPBM) plates, and growth on acetate (YPBA) as a control, in triplicate. One record is one deletion strain on one carbon source with its ordinal score.	4,770	3	14,310	environment_response (ordinal)	scalar	global	1.8×10^{6}	good	1 gene x condition
HetHillenmeyer2008	*S. cerevisiae*	Hillenmeyer 2008 (FitDb) grew the barcoded heterozygous (HIP) deletion collection, one allele replaced by kanMX4 in BY4743, competitively under chemical and environmental conditions read on barcode arrays. This store holds the HIP fitness-defect log2 ratio against matched control arrays: one record is one constructed strain on one treatment array.	17,659	512	3,074,132	environment_response (log2_ratio)	scalar	global	2.0×10^{9}	good	1 gene x condition
HomHillenmeyer2008	*S. cerevisiae*	Hillenmeyer 2008 (FitDb) grew the barcoded homozygous (HOP) kanMX4 deletion collection in BY4743 competitively under chemical and environmental conditions read on barcode arrays. This store holds the HOP fitness-defect z-score against a matched control array set, since no HOM log ratio was archived: one record is one constructed strain on one treatment array.	9,220	276	1,228,676	environment_response (z_score)	scalar	global	8.4×10^{8}	good	1 gene x condition
IsolateGrowthPeter2018	*S. cerevisiae*	Peter 2018 phenotyped 971 sequenced natural *S. cerevisiae* isolates by end-point colony growth on solid medium under 35 stress, carbon-source and temperature conditions. One record is one isolate on one condition: its colony size divided by its own colony size on YPD at 30 C, with the isolate genotype built as sequence variants and gene presence off S288C (33,985 records).	971	35	33,985	environment_response (condition_growth_ratio)	scalar	global	1.6×10^{7}	no	natural genome (segregant or isolate)
Expression (RNA-seq)											
Boocock2025	*S. cerevisiae*	Boocock 2025 profiled four segregant panels from Bloom 2019 crosses by 10x 3' single-cell RNA-seq, genotyping each cell from its transcribed SNPs. One record is either a pooled, previously genotyped BY x RM segregant (its cells summed into pseudobulk UMI counts) or one de novo segregant cell with its own counts and haplotype posterior.	79,516	2	79,516	umi_count (scrnaseq_umi_count)	vector (3797)	node	8.9×10^{9}	no	natural genome (segregant or isolate)
CaudalPanTranscriptome2024	*S. cerevisiae*	Caudal 2024 measured whole-transcriptome expression by RNA-seq in natural *S. cerevisiae* isolates of the 1002-genomes panel. Each of the 943 isolates with a Peter 2018 genome is a perturbation set off S288C (allele variants, gene presence, absence, loss of function). One record is one isolate with per-gene TPM and raw counts.	943	1	943	expression_tpm (rnaseq_tpm)	vector (6017)	node	1.7×10^{9}	no	natural genome (segregant or isolate)
NadalRibellesPerturbSeq2025	*S. cerevisiae*	Nadal-Ribelles 2025 rebuilt the yeast knockout collection with RNA-readable genotype barcodes (KanMX4 replaced by URA3) and profiled over 3,500 nonessential deletions by microwell single-cell RNA-seq in YPD and in 0.4 M NaCl osmostress. One record is one barcoded genotype in one condition: pseudobulk log2 fold change versus wild type, cell-to-cell dispersion and cell count.	3,150	2	6,188	expression_log2_ratio (pseudobulk_scrnaseq_log2fc)	vector (5818)	node	2.5×10^{9}	best	1 gene, its own label x condition
Nguessan2025	*S. cerevisiae*	N'Guessan 2025 grew 4,489 F2 segregants of a BY4741 x RM11-1a cross together in YPD and profiled them in one 10x 3' single-cell RNA-seq run, assigning each cell to a segregant by its transcribed SNPs. One record is one segregant (2,951 in all): the summed UMI counts of its assigned cells, with its genotype as RM-allele posteriors at 41,594 SNPs.	2,951	1	2,951	umi_count (scrnaseq_umi_count)	vector (1670)	node	4.5×10^{8}	no	natural genome (segregant or isolate)
PooledPromoterExpressionVaishnav2022	*S. cerevisiae*	Vaishnav 2022 measured expression of 71,158 promoter sequences by a pooled sort-seq reporter assay: an 80 bp variable insert in a 110 bp scaffold drives YFP from a low-copy CEN plasmid in *S. cerevisiae* grown in SC-Ura. One record is one library sequence: the sequence as genotype and its estimated expression level (104 released as NA).	71,158	1	71,158	expression_level	scalar	node	1.3×10^{8}	best	inserted DNA (promoter or reporter)
TfDeletionPseudobulkJackson2020	*S. cerevisiae*	Jackson 2020 pooled 72 barcoded prototrophic diploid strains, six independent constructions each of eleven nitrogen-regulatory transcription-factor deletions and an HO-deletion control, grew them in eleven conditions and profiled them by 10x single-cell RNA-seq. One record is one strain in one condition: summed UMI counts of its cells (784 records).	72	11	784	umi_count (scrnaseq_umi_count)	vector (5075)	node	9.7×10^{7}	best	1 gene, its own label x condition
Expression (microarray)											
DmMicroarraySameith2015	*S. cerevisiae*	Sameith 2015 profiled double-deletion strains pairing general stress transcription factors (GSTFs) on two-channel microarrays against a common wild-type reference pool, each grown and profiled four times from two independent cultures. One record is one of 72 GSTF pairs with log2(mutant/reference) ratios over 6,169 genes.	72	1	72	expression_log2_ratio	vector (6169)	node	1.7×10^{8}	good	readout gene x 2 perturbed gene(s)
InductionTimecourseHackett2020	*S. cerevisiae*	Hackett 2020 placed each of about 200 *S. cerevisiae* regulators under a beta-estradiol-inducible synthetic promoter, grew each strain to steady state in a chemostat, induced with 1 uM beta-estradiol and profiled RNA on two-color Agilent arrays over time. One record is one array: one time course at one time point, as cleaned log2 ratio versus the same culture at t = 0.	214	31	1,752	log2_cleaned	vector (6175)	node	4.7×10^{9}	best	1 gene, its own label x condition (0-1 edits)
MicroarrayKemmeren2014	*S. cerevisiae*	Kemmeren 2014 profiled single-gene deletion mutants of *S. cerevisiae* (1,484 in its Table S1) on two-color microarrays against a common wild-type reference RNA pool, two independent cultures each hybridized in dye-swap for four measurements per mutant. One record is one deletion strain: per-gene log2 ratio versus the reference pool, stored as liquid SC at 30 C.	1,484	1	1,484	expression_log2_ratio	vector (6169)	node	3.2×10^{9}	best	1 gene, its own label
MicroarrayScoreHu2007	*S. cerevisiae*	Hu 2007 profiled *S. cerevisiae* transcription-factor deletion strains (BY4741, KanMX) grown to mid-log in YPD against a wild-type reference on two-color spotted cDNA arrays (GEO GSE4654). One record is one deletion strain: the released error-model X score and its p-value for each of 6,429 array elements, averaged over at least two replicate arrays.	263	1	263	expression_score (error_model_x_score_weighted_duplicate_average)	vector (6371)	node	2.5×10^{8}	best	1 gene, its own label
SmMicroarraySameith2015	*S. cerevisiae*	Sameith 2015 profiled 82 single deletions of general stress transcription factors (BY4742, MATalpha, KanMX) on two-color microarrays against a common wild-type RNA reference in dye-swap, two cultures each for four measurements. One record is one single-deletion strain: per-gene log2 ratio versus the reference, stored as liquid SC at 30 C.	82	1	82	expression_log2_ratio	vector (6169)	node	1.8×10^{8}	best	1 gene, its own label
Fitness											
Bloom2019	*S. cerevisiae*	Bloom 2019 phenotyped 13,950 haploid segregants from 16 biparental crosses by colony radius on solid plates in 38 growth conditions (carbon sources, stress compounds, pH, temperature). Each segregant is a haplotype mosaic of its two parents. One record is one segregant in one condition, mostly the radius residual against its matched control plate.	13,950	38	530,100	environment_response (control_regression_residual)	scalar	global	2.6×10^{8}	no	natural genome (segregant or isolate)
CrispriSegregantHale2024	*S. cerevisiae*	Hale 2024 integrated a dCas9-Mxi1 CRISPRi guide library into 169 haploid segregants of a BY4716 x 322134S cross and the BY parent, and measured lineage fitness by double-barcode sequencing in two ATc-induced assays and one uninduced assay. One record is one double-barcode lineage in one assay, with its relative fitness and standard error.	895,494	2	8,692,635	environment_response (relative_fitness)	scalar	global	6.7×10^{9}	good	1 gene x condition
DmfCostanzo2016	*S. cerevisiae*	Costanzo 2016 crossed query mutants (deletions and temperature-sensitive alleles) to deletion and ts arrays by synthetic genetic array (SGA) and measured colony size at 26 or 30 C, four replicate colonies per screen. This store holds double-mutant fitness: one record is one query x array strain pair, 20,705,612 in all, with its SD.	20,705,612	2	20,705,612	fitness	scalar	global	7.1×10^{9}	good	gene pair, strain readout x condition
DmfKuzmin2018	*S. cerevisiae*	Kuzmin 2018 ran SGA screens at 26 C crossing query strains (single mutants marked with an ho deletion, or double mutants) to deletion and ts-allele arrays, scoring colony size. This store holds double-mutant fitness: 410,399 digenic query x array crosses plus 172 double-mutant query strains of the trigenic screens, each record one strain with its SD.	410,571	1	410,571	fitness	scalar	global	1.5×10^{8}	good	gene pair, strain readout
DmfKuzmin2020	*S. cerevisiae*	Kuzmin 2020 ran SGA screens at 26 C crossing ho-marked single and double mutant query strains to deletion and ts-allele arrays, including pilot screens (Table S3). This store holds double-mutant fitness: 632,797 digenic query x array crosses plus 201 double-mutant query strains from Table S5, each record one strain with its fitness SD.	608,805	1	632,998	fitness	scalar	global	2.4×10^{8}	good	gene pair, strain readout
SmfBaryshnikova2010	*S. cerevisiae*	Baryshnikova 2010 estimated single-mutant fitness from SGA colony sizes on control screens for 6,023 *S. cerevisiae* alleles: 4,635 nonessential kanMX deletions, 1,082 essential-gene DAmP hypomorphs and 306 temperature-sensitive alleles. One record is one allele: fitness normalized to a mode of 1.0 with a bootstrap standard error.	6,023	2	6,023	fitness	scalar	global	1.8×10^{6}	good	1 gene x condition
SmfCostanzo2016	*S. cerevisiae*	Costanzo 2016 estimated single-mutant fitness of *S. cerevisiae* deletion, DAmP, temperature-sensitive and other alleles from SGA colony sizes averaged over replicate control screens at 26 C and 30 C on solid SGA selection medium. One record is one strain at one temperature: fitness relative to wild type with a bootstrap standard error.	10,277	2	20,484	fitness	scalar	global	5.4×10^{6}	good	1 gene x condition
SmfKuzmin2018	*S. cerevisiae*	Kuzmin 2018 released single-mutant fitness from its SGA trigenic screen: array strains (kanMX deletions and temperature-sensitive alleles) and digenic-screen query strains, scored by colony size at 26 C on solid triple-mutant selection medium. One record is one array or query single mutant, with query SDs from Additional Data S4.	1,539	1	1,539	fitness	scalar	global	3.9×10^{5}	best	1 gene
SmfKuzmin2020	*S. cerevisiae*	Kuzmin 2020 estimated fitness of single-mutant query strains (kanMX deletions and other alleles) from colony sizes on a high-density array, 12 to 24 colony measurements per estimate, at 26 C on solid SGA triple-mutant selection medium. One record is one single-mutant query strain: fitness and bootstrap SD from Table S5.	472	1	472	fitness	scalar	global	9.3×10^{4}	best	1 gene
SmfODuibhir2014	*S. cerevisiae*	O'Duibhir 2014 measured doubling times of 1,312 *S. cerevisiae* gene-deletion strains from the OD600 growth slope in liquid synthetic complete medium with 2% glucose, averaging two biological replicate cultures. One record is one deletion strain: fitness as growth rate relative to wild type, 2 to the power of minus the released log2 relative doubling time.	1,312	1	1,312	fitness	scalar	global	2.0×10^{6}	best	1 gene
TmfKuzmin2018	*S. cerevisiae*	Kuzmin 2018 measured triple-mutant fitness by SGA, crossing double-mutant query strains against an array of kanMX deletion and temperature-sensitive alleles and scoring colony size at 26 C on solid triple-mutant selection medium. One record is one trigenic query x array cross: combined-mutant fitness with a sample SD over colonies (Data S1).	91,111	1	91,111	fitness	scalar	global	3.9×10^{7}	good	gene triple, strain readout
TmfKuzmin2020	*S. cerevisiae*	Kuzmin 2020 measured triple-mutant fitness by SGA, crossing 240 double-mutant query strains against about 1,200 array strains (Table S1) and against the genome-wide nonessential and temperature-sensitive arrays in pilot screens (Table S3), at 26 C. One record is one trigenic query x array cross in one screen: fitness and colony SD.	290,355	1	301,798	fitness	scalar	global	1.3×10^{8}	good	gene triple, strain readout
Gene interaction											
DmiCostanzo2016	*S. cerevisiae*	Costanzo 2016 crossed query mutants (deletions and temperature-sensitive alleles) to deletion and ts arrays by synthetic genetic array (SGA) at 26 or 30 C, scoring colony size. This store holds the digenic genetic interaction score (epsilon) and its P-value: one record is one query x array gene pair, 20,705,612 records across four Data File S1 tables.	20,705,612	2	20,705,612	gene_interaction	scalar	edge	6.0×10^{9}	best	gene pair
DmiKuzmin2018	*S. cerevisiae*	Kuzmin 2018 ran SGA screens at 26 C crossing ho-marked query strains to deletion and ts-allele arrays, scoring colony size. This store holds the digenic genetic interaction score (epsilon) and its P-value from the raw data table: one record is one query x array gene pair, 410,399 records in all.	410,399	1	410,399	gene_interaction	scalar	edge	1.3×10^{8}	best	gene pair
DmiKuzmin2020	*S. cerevisiae*	Kuzmin 2020 ran SGA screens at 26 C crossing ho-marked query strains to deletion and ts-allele arrays, in a main screen (Table S1) and pilot screens (Table S3). This store holds the digenic genetic interaction score (epsilon) and its P-value: one record is one query x array gene pair, 632,797 records across both tables.	608,604	1	632,797	gene_interaction	scalar	edge	2.1×10^{8}	best	gene pair
TmiKuzmin2018	*S. cerevisiae*	Kuzmin 2018 scored trigenic genetic interactions from SGA colony sizes of double-mutant query strains crossed against an array of deletion and temperature-sensitive alleles at 26 C. One record is one gene triple from a trigenic query x array cross: the adjusted interaction score tau and its P-value, with zero as the reference (Data S1).	91,111	1	91,111	gene_interaction	scalar	hyperedge	3.4×10^{7}	best	gene triple
TmiKuzmin2020	*S. cerevisiae*	Kuzmin 2020 scored trigenic interactions from SGA screens of 240 double-mutant query strains against about 1,200 array strains (Table S1) and against genome-wide arrays in pilot screens (Table S3), at 26 C. One record is one gene triple from one screen's trigenic query x array cross: the adjusted interaction score tau and its P-value.	290,355	1	301,798	gene_interaction	scalar	hyperedge	1.1×10^{8}	best	gene triple
Genome architecture											
GenomeArchitecturePuddu2019	*S. cerevisiae*	Puddu 2019 whole-genome sequenced colonies of the homozygous diploid yeast knockout collection and released copy numbers of rDNA, CUP1, mtDNA, 2 micron and five Ty families, a telomere-length proxy, all 16 chromosomes and a rearrangement count. One record is one sequenced knockout colony, 8,835 over 4,458 deletion strains.	4,550	1	8,835	genome_architecture	vector (28)	global	1.2×10^{7}	best	1 gene
Life cycle											
LifeCycleDeChiara2022	*S. cerevisiae*	De Chiara 2022 phenotyped the 1,011 sequenced natural *S. cerevisiae* isolates for the sexual and aging parts of the life cycle. One record is one isolate under one assay protocol, a time course: sporulation efficiency and dyad fraction in potassium acetate (24 h, 72 h), sporulation in PBS, or stationary-phase viability in PBS, SDC or calorie-restricted SDC.	905	5	4,039	life_cycle	vector (4)	global	1.3×10^{6}	no	natural genome (segregant or isolate)
Metabolite											
AminoAcidCooper2010	*S. cerevisiae*	Cooper 2010 measured free amine-containing metabolite pools of the haploid MATa deletion collection after overnight growth in synthetic complete medium, by NBD-F derivatization and capillary electrophoresis with laser-induced fluorescence. One record is one deletion strain with 17 peak values, each a linear ratio to its plate average.	4,334	1	4,382	metabolite_level (ce_lif_peak_area_ratio_to_plate_mean)	vector (16)	bipartite node	5.2×10^{6}	good	metabolite x 1 perturbed gene(s)
AminoAcidMulleder2016	*S. cerevisiae*	Mulleder 2016 measured the intracellular concentration of 19 amino acids by LC-SRM/MS in 4,678 strains of the prototrophic yeast deletion collection grown exponentially on synthetic minimal agar. One record is one deletion strain with 19 batch-normalized concentrations in mM, against the population robust mean as reference.	4,678	1	4,678	metabolite_level (intracellular_concentration_mM)	vector (19)	bipartite node	1.1×10^{7}	good	metabolite x 1 perturbed gene(s)
BetaxanthinCachera2023	*S. cerevisiae*	Cachera 2023 used CRI-SPA to transfer four betaxanthin biosynthesis genes into each strain of the yeast knockout (YKO) collection on YPD with G418, then scored colony yellowness by image analysis as a betaxanthin proxy. One record is one gene deletion with its 24 h corrected yellowness score, SE and replicate count.	4,759	1	4,759	metabolite_level (cri_spa_corrected_hsv_yellowness_24h)	scalar	bipartite node	4.5×10^{6}	good	metabolite x 5 perturbed gene(s)
FattyAcidXue2025	*S. cerevisiae*	Xue 2025 is an in-house screen of a free fatty acid chassis (BY4741 with POX1, FAA1 and FAA4 deleted) carrying one to three further transcription factor deletions from 10 TFs, assayed by GC for absolute titer. One record is one genotype with mg/L titers of five FFAs (C14:0, C16:0, C18:0, C16:1, C18:1), mean and SE of three replicates.	176	1	176	metabolite_level (titer_mg_per_l)	vector (5)	bipartite node	1.9×10^{5}	good	metabolite x 6 perturbed gene(s) (3-6 edits)
IsobutanolScreenLopez2024	*S. cerevisiae*	Montaño López 2024 integrated a GFP isobutanol-pathway biosensor under the LEU1 promoter into every strain of the BY4741 yeast knockout collection and read median fluorescence by flow cytometry in liquid SC medium. This first genome-wide screen measured each strain once; one record is one released row, the fold change over the same-plate wild-type control (4,805 rows).	4,607	1	4,805	metabolite_level (biosensor_gfp_fluorescence_fold_change)	scalar	bipartite node	9.0×10^{5}	good	metabolite x 1 perturbed gene(s)
IsobutanolValidatedLopez2024	*S. cerevisiae*	Montaño López 2024 re-screened the strong hits of the isobutanol-biosensor knockout screen (66 up at fold change 2 or more, 161 down at 0.5 or less) in triplicate by flow cytometry in liquid SC medium. One record is one deletion strain: its median GFP fluorescence fold change over the same-plate wild type, with SE from the released sample SD over three replicates (Table S3).	226	1	227	metabolite_level (biosensor_gfp_fluorescence_fold_change)	scalar	bipartite node	7.4×10^{4}	good	metabolite x 1 perturbed gene(s)
MetaboliteDaSilveira2014	*S. cerevisiae*	da Silveira dos Santos 2014 measured lipidomes of 127 nonessential protein kinase and phosphatase deletion mutants by mass spectrometry (MRM-MS and Orbitrap for 147 acyl-chain-resolved glycero- and sphingolipid species, GC-MS for ergosterol). One record is one mutant: relative lipid abundances in arbitrary units, referenced to the mean of three measured wild-type strains.	127	1	127	metabolite_level (lipidomics_ms_relative_abundance_au)	vector (135)	bipartite node	1.1×10^{6}	good	metabolite x 1 perturbed gene(s)
MetaboliteZelezniak2018	*S. cerevisiae*	Zelezniak 2018 measured the targeted central-carbon and amino-acid metabolome of 95 kinase-deletion strains of *S. cerevisiae* plus wild type by LC-SRM under three protocols with different units and cultures. One record is one strain under one protocol: per-metabolite mean and standard error over replicates, referenced to wild type measured under the same protocol.	95	1	129	metabolite_level (lc_srm_signal_batch_corrected_uncalibrated_dataset1)	vector (14)	bipartite node	3.5×10^{5}	good	metabolite x 1 perturbed gene(s)
OrganicAcidYoshida2012	*S. cerevisiae*	Yoshida 2012 screened the *S. cerevisiae* haploid deletion collection for organic-acid overproducers and quantified 17 gene deletions (KanMX4) by HPLC. One record is one deletion strain: titers in mM of acetate, citrate, malate, pyruvate, succinate and phosphate, mean of three independent experiments, referenced to the measured BY4742 wild type (Table 3).	17	1	17	metabolite_level (hplc_organic_acid_titer_mM)	vector (6)	bipartite node	3.3×10^{4}	good	metabolite x 1 perturbed gene(s)
Morphology											
CarotenoidOzaydin2013	*S. cerevisiae*	Ozaydin 2013 transformed the haploid BY4741 deletion collection with the carotenogenic plasmid YB/I/BTS1 (crtYB and crtI plus an extra BTS1 copy) and visually scored colony color, a beta-carotene proxy, on a -5 to +5 scale with wild type at 0. One record is one released plate row: a deletion strain and its color score.	4,475	1	4,607	visual_score	scalar	global	7.7×10^{6}	weak	set of 4 edits incl. heterologous genes
ScmdOhnuki2018	*S. cerevisiae*	Ohnuki 2018 imaged heterozygous diploid deletions of 1,112 essential genes in BY4743 (one of two copies removed, kanMX4) grown in liquid YPD at 25 C, and quantified single-cell morphology with CalMorph. One record is one strain: a 501-trait vector of population averages and variation parameters, referenced to the mean of 114 wild-type replicates.	1,112	1	1,112	calmorph	vector (281)	global	4.9×10^{7}	best	1 gene
ScmdOhnuki2022	*S. cerevisiae*	Ohnuki 2022 deleted each target gene in a drug-hypersensitive pdr1Δ pdr3Δ snq2Δ background, grew the quadruple deletions in liquid YPD at 25 C, triple-stained and imaged them, and quantified morphology with CalMorph. One record is one quadruple-deletion strain (1,979): a 501-trait vector, referenced to the mean of 749 replicates of the triple-deletion parent.	1,982	1	1,982	calmorph	vector (281)	global	8.8×10^{7}	no	multi-edit strain, 4 edits
ScmdOhya2005	*S. cerevisiae*	Ohya 2005 imaged the haploid BY4741 nonessential single-gene deletion collection (kanMX4) grown in YPD and quantified single-cell morphology of fixed log-phase cells with CalMorph. One record is one full deletion strain (4,718 in the matrix): a 501-trait vector of CalMorph averages and variation parameters, referenced to the mean of 122 wild-type replicates.	4,718	1	4,718	calmorph	vector (281)	global	1.5×10^{8}	best	1 gene
Protein abundance											
ProteomeMessner2021	*S. cerevisiae*	Messner 2021 measured proteomes of one prototrophic BY4741 wild-type strain grown in synthetic minimal medium and treated overnight with 16 drugs (statins, azoles, antifolates), folic acid or FIN56 at 10 uM, by Scanning SWATH mass spectrometry. One record is one drug-treated run: per-protein DIA-NN MaxLFQ quantities, referenced to 24 DMSO runs (70 records).	1	18	70	protein_abundance (swath_ms_diann_maxlfq_proteotypic_quantity)	vector (1920)	node	1.2×10^{7}	best	1 gene, unperturbed strain x condition
ProteomeMessner2023	*S. cerevisiae*	Messner 2023 measured proteomes of the prototrophic *S. cerevisiae* haploid gene-deletion collection by microflow SWATH-MS, each strain once, in synthetic minimal medium. One record is one knockout sample (4,699 strains, duplicate strains of one ORF kept apart): non-imputed MaxLFQ protein quantities, referenced to 388 replicates of a his3Δ HIS3-complemented control.	4,549	1	4,699	protein_abundance (swath_ms_maxlfq_batch_corrected_quantity)	vector (1830)	node	1.2×10^{9}	best	1 gene, its own label
ProteomeMuenzner2024	*S. cerevisiae*	Muenzner 2024 measured proteomes of natural *S. cerevisiae* isolates of the 1,011-genomes panel by SWATH-MS (DIA-NN) in synthetic minimal medium. One record is one strain (794 isolates with a Peter genome plus FY4): released non-imputed, batch-corrected MaxLFQ quantities per protein, with each isolate genotype built off S288C (SI Table 3).	795	1	795	protein_abundance (swath_ms_maxlfq_precursor_batch_corrected_quantity)	vector (1542)	node	9.6×10^{8}	no	natural genome (segregant or isolate)
ProteomeZelezniak2018	*S. cerevisiae*	Zelezniak 2018 measured the proteome of 97 kinase-deletion strains of *S. cerevisiae* plus wild type by SWATH-MS label-free quantification of 726 proteins. One record is one knockout strain: per-protein mean and standard error of batch-corrected log signal over at least two replicate samples, referenced to the wild-type parent.	97	1	97	protein_abundance (swath_ms_label_free_log_signal_sva)	vector (726)	node	1.7×10^{7}	best	1 gene, its own label
Protein synthesis / turnover											
ProteinTurnoverMuenzner2024	*S. cerevisiae*	Muenzner 2024 measured protein turnover in 55 natural *S. cerevisiae* isolates of the 1,011-genomes panel by dynamic SILAC in synthetic minimal medium. One record is one isolate: per-protein turnover rate kdp (1/h) and half-life (h), computed without growth-dilution correction, so the rate is total replacement (SI Table 17).	55	1	55	degradation_rate (dynamic_silac_lys8_kdp_per_hour_no_dilution_correction)	vector (1015)	node	6.9×10^{7}	no	natural genome (segregant or isolate)
Viability											
GeneEssentialitySgd	*S. cerevisiae*	Built from SGD per-gene phenotype annotations: every annotation with mutant type null, strain S288C and phenotype inviable is one record. The allele is a whole-gene deletion or, for 42 annotations, a named point, truncation or internal-deletion allele, and any curated condition is kept. Each record carries the SGD experiment type as essentiality evidence.	1,180	5	1,329	is_essential	scalar	node	6.8×10^{5}	best	1 gene, its own label x condition
SynthLethalityYeastSynthLethDb	*S. cerevisiae*	SynLethDB 2.0 aggregates *S. cerevisiae* synthetic lethal gene pairs from literature and databases such as BioGRID, each with its source, PubMed id and a normalized confidence score, but no medium or temperature. One record is one released gene pair (14,000), each gene stored by its released name and NCBI Entrez id, with the pair marked synthetic lethal.	14,000	1	14,000	is_synthetic_lethal	scalar	edge	8.0×10^{6}	best	gene pair
SynthRescueYeastSynthLethDb	*S. cerevisiae*	SynLethDB 2.0 also collects *S. cerevisiae* synthetic rescue gene pairs from the same literature and database sources, with source and PubMed id but no medium, temperature or defined score. One record is one released gene pair (6,948), each gene stored by its released name and NCBI Entrez id, with the pair marked synthetic rescue.	6,948	1	6,948	is_synthetic_rescue	scalar	edge	4.0×10^{6}	best	gene pair
Total (69 datasets, 69 probed)					**93,266,786**				7.2×10^{10}		